SIMATS ENGINEERING

ASSESSMENT TOOL III – VISUALIZATION CRITIQUE & REDESIGN

Course: Data Handling and Visualization	Code: DSA0601	CO: CO3
Duration: 120 min	Total Marks: 100	Reg No: 192424303
Name: P.Sunayana		

COURSE OUTCOME COVERED

CO3: Analyze time-series data, trends, associations, and uncertainty through appropriate visualization methods. (BL4) Activity Tool: Visualization Critique & Redesign Weightage: 20%

Code of Conduct:

I **__P.Sunayana(192424303)__** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: __P.Sunayana__

Criteria	Evaluation of Existing Visualization (25%)	Redesign Strategy (25%)	Application of Vis. Principles (20%)	Organization (15%)	Accuracy (10%)	Clarity (5%)	Total
Q1							
Q2							
Q3							
Q4							
Q5							

AT3 – Visualization Critique & Redesign

R Programming Assessment – Distribution and Proportion Visualization

Instructions: For each question, use R/RStudio to reproduce the given visualization, critique its effectiveness, identify misleading or inappropriate visual encodings, and redesign it using a more suitable visualization. Include the R code, redesigned visualization, and a concise justification of your design decisions.

Question 1 – Critique and Redesign an ECDF and Q-Q Plot

Dataset: Delivery_Time_Distribution.csv

Dataset:

Order_ID	Warehouse	Delivery_Time_Hrs
O01	Chennai	4
O02	Chennai	5
O03	Chennai	6
O04	Chennai	7
O05	Chennai	9
O06	Bengaluru	3
O07	Bengaluru	5
O08	Bengaluru	6
O09	Bengaluru	8
O10	Bengaluru	10
O11	Hyderabad	6

O12	Hyderabad	7
O13	Hyderabad	8
O14	Hyderabad	12
O15	Hyderabad	15
O16	Pune	4
O17	Pune	6
O18	Pune	8
O19	Pune	9
O20	Pune	22

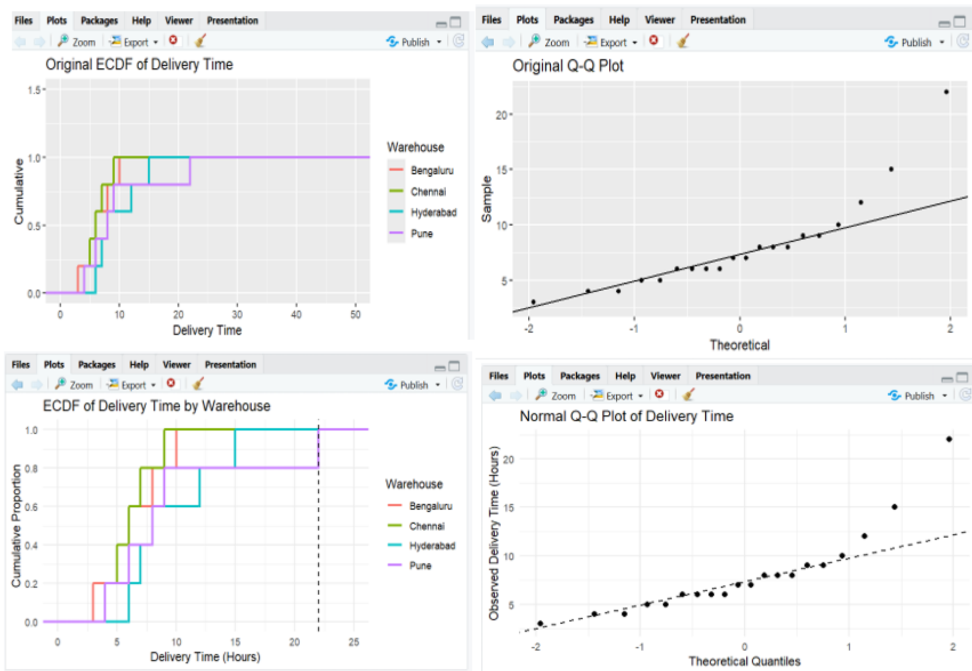
Assessment Task

A logistics analyst has created an ECDF and a Q-Q plot to study delivery time distribution across warehouses, but the charts use inconsistent axis scaling, missing reference lines, and cluttered gridlines.

1. Create the two original-style plots in R.

2. Identify at least three visualization problems.

3. Redesign the plots using appropriate scales, labels, reference lines, and minimal visual clutter. 4. Explain how the redesigned plots improve interpretation of skewness, outliers, and normality. 5. State whether an ECDF plot alone is sufficient to detect outliers, or whether it should be paired with another plot.



ECDF and Q-Q Plot

- ECDF shows the cumulative distribution of delivery times.
- Q-Q plot checks whether the data follows a normal distribution.
- Original plots had inconsistent axis scaling and unnecessary clutter.
- Redesigned plots use proper scales, clear labels, and reference lines.
- The 22-hour delivery time appears to be a possible outlier.
- The Q-Q plot helps identify skewness and deviations from normality.
- ECDF alone is not sufficient for detecting outliers.
- It should be combined with a boxplot or Q-Q plot.

Question 2 – Critique and Redesign Many Distributions Using Bean Plots

Dataset: Subject_Marks.csv

Dataset:

Student_ID	Subject	Marks
T01	Physics	58
T02	Physics	63

T03	Physics	70
T04	Physics	77
T05	Physics	85
T06	Chemistry	52
T07	Chemistry	60
T08	Chemistry	68
T09	Chemistry	74
T10	Chemistry	82
T11	Mathematics	61
T12	Mathematics	66
T13	Mathematics	73
T14	Mathematics	80
T15	Mathematics	90
T16	Biology	55
T17	Biology	62
T18	Biology	69
T19	Biology	76
T20	Biology	88

Assessment Task

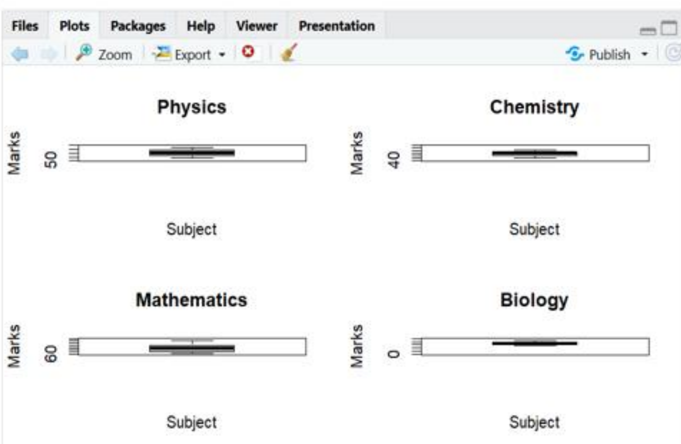
An academic report uses four separate boxplots with different y-axis ranges to compare marks across subjects. The comparison is difficult because the scales and layouts are inconsistent.

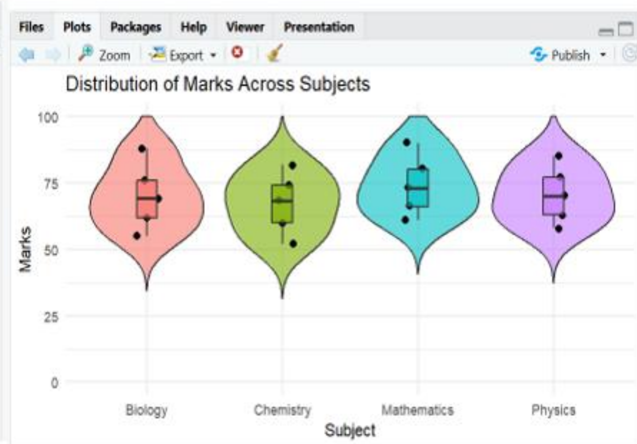
1. Reproduce the problematic visualization using R.

2. Critique the use of separate boxplots, inconsistent scales, and truncated axes.

3. Redesign the visualization using bean plots or a suitable violin/density-based alternative. 4. Add an appropriate common scale and clear subject labels.

5. Explain which subject has the highest concentration of marks and which subject shows the greatest spread.





Bean/Violin Plot

- Original visualization used separate boxplots with different scales.
- Different y-axis ranges make comparisons misleading.
- Truncated axes can exaggerate differences.
- A combined violin/bean plot uses a **common scale from 0 to 100**.
- All subject distributions can be compared easily in one chart.
- Individual data points can also be displayed.
- **Mathematics shows the greatest spread** in marks.
- The redesigned visualization is clearer and more compact.

Question 3 – Critique and Redesign Proportion Visualizations

Dataset: App_Downloads.csv

Dataset:

City	**App_Category**	**Downloads**
Chennai	Games	30
Chennai	Social	42
Chennai	Utility	18
Mumbai	Games	35
Mumbai	Social	38
Mumbai	Utility	22
Delhi	Games	26
Delhi	Social	44
Delhi	Utility	20
Kolkata	Games	24
Kolkata	Social	36
Kolkata	Utility	29

Assessment Task

A marketing team presents app-category proportions using four pie charts, one for each city. The charts use exploded slices, mismatched color palettes, and inconsistent labeling.

1. Create a representative version of the misleading visualization.

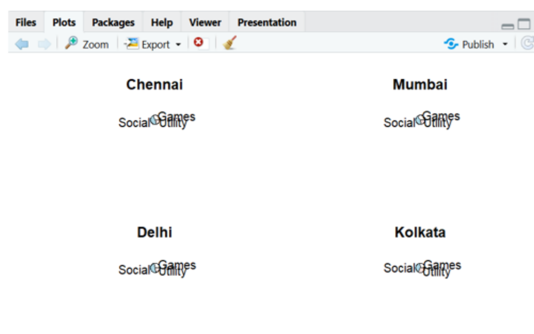
2. Identify at least four design problems.

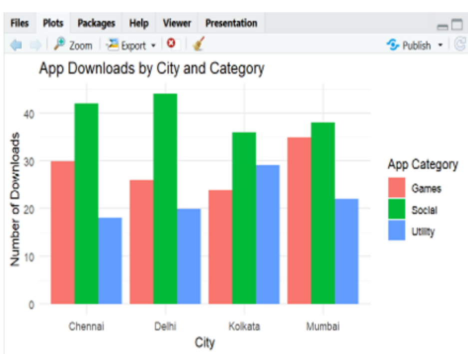
3. Redesign the analysis using side-by-side bars and a 100% stacked bar chart.

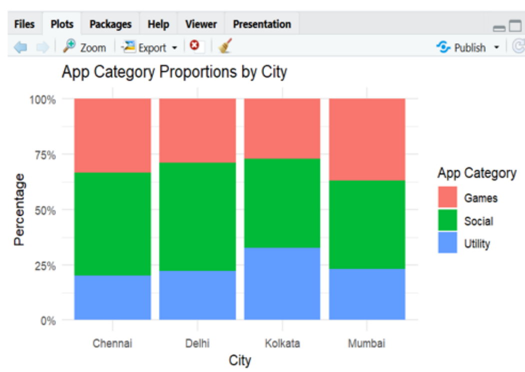
4. Compare the two redesigns and determine which is better for comparing city-wise proportions.

5. Explain why exploded pie slices and mismatched color palettes can lead to misleading conclusions.

```
      City App_Category Downloads
1  Chennai        Games        30
2  Chennai       Social        42
3  Chennai      Utility        18
4   Mumbai        Games        35
5   Mumbai       Social        38
6   Mumbai      Utility        22
7    Delhi        Games        26
8    Delhi       Social        44
9    Delhi      Utility        20
10 Kolkata        Games        24
11 Kolkata       Social        36
12 Kolkata      Utility        29
```







Proportion Visualization

- Original visualization used separate pie charts for each city.
- Pie slices are difficult to compare across multiple charts.

- Exploded slices can exaggerate the importance of categories.
- Different colors for the same category can confuse users.
- Side-by-side bars are useful for comparing actual downloads.
- A 100% stacked bar chart is useful for comparing proportions.
- The 100% stacked bar chart is better for city-wise proportion comparison.
- Consistent colors improve interpretation.

Question 4 – Critique and Redesign Nested Proportions

Dataset: Campus_Expenditure.csv

Dataset:

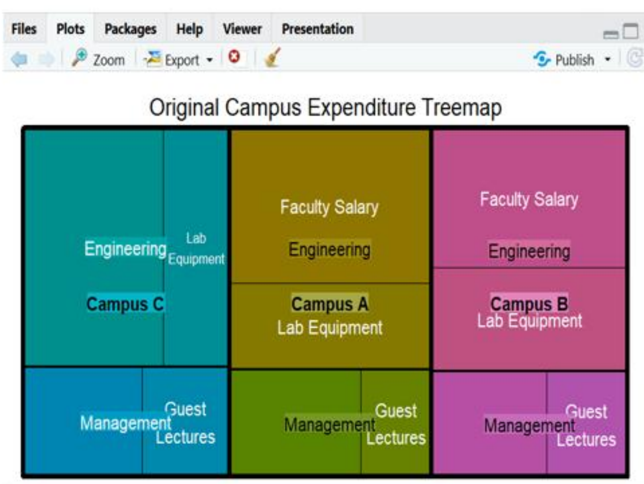
Campus	Department	Expense_Type	Amount
Campus A	Engineering	Faculty Salary	62000
Campus A	Engineering	Lab Equipment	34000
Campus A	Management	Faculty Salary	28000
Campus A	Management	Guest Lectures	15000
Campus B	Engineering	Faculty Salary	54000
Campus B	Engineering	Lab Equipment	40000
Campus B	Management	Faculty Salary	24000
Campus B	Management	Guest Lectures	17000
Campus C	Engineering	Faculty Salary	66000

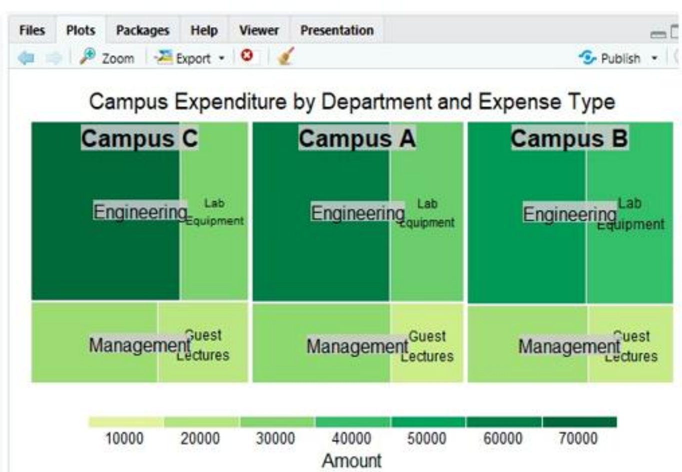
Campus C	Engineering	Lab Equipment	31000
Campus C	Management	Faculty Salary	26000
Campus C	Management	Guest Lectures	19000

Assessment Task

A finance office uses a treemap to communicate Campus → Department → Expense Type spending. Small areas have unreadable labels and the color scale does not clearly communicate the hierarchy.

1. Create the original treemap.
2. Critique the size, labeling, color, and hierarchy encodings.
3. Redesign the visualization as an improved treemap.
4. Create a sunburst alternative and compare it with the redesigned treemap.
5. Recommend one visualization for a board presentation and justify your choice.





Nested Proportions

- The data follows the hierarchy: **Campus → Department → Expense Type**.
- The original treemap may contain small areas with unreadable labels.
- Small rectangles are difficult to compare accurately.

- The color scheme may not clearly show the hierarchy.
- The improved treemap uses clearer labels and boundaries.
- A sunburst chart can also represent the hierarchical structure.
- The sunburst is better for understanding relationships between levels.
- The **treemap is better for comparing expenditure amounts**.
- Treemap is recommended for a board presentation.

Question 5 – Critique and Redesign Flow-Based Proportions

Dataset: Course_to_Career_Flow.csv

Dataset:

Course	Certification	Job_Role	Students
Data Science	Yes	Analyst	22
Data Science	Yes	Engineer	14
Data Science	No	Analyst	9
Web Development	Yes	Frontend Dev	19
Web Development	Yes	Backend Dev	11
Web Development	No	Frontend Dev	7
Cloud Computing	Yes	DevOps Engineer	17
Cloud Computing	Yes	Cloud Analyst	10
Cloud Computing	No	Support Engineer	6
UI/UX Design	Yes	Product Designer	13
UI/UX Design	Yes	UX Researcher	8

UI/UX Design	No	Product Designer	5

Assessment Task

A training institute uses a Sankey diagram to show Course → Certification → Job Role. The original chart uses near identical colors, unlabelled flows, and node widths that are difficult to compare.

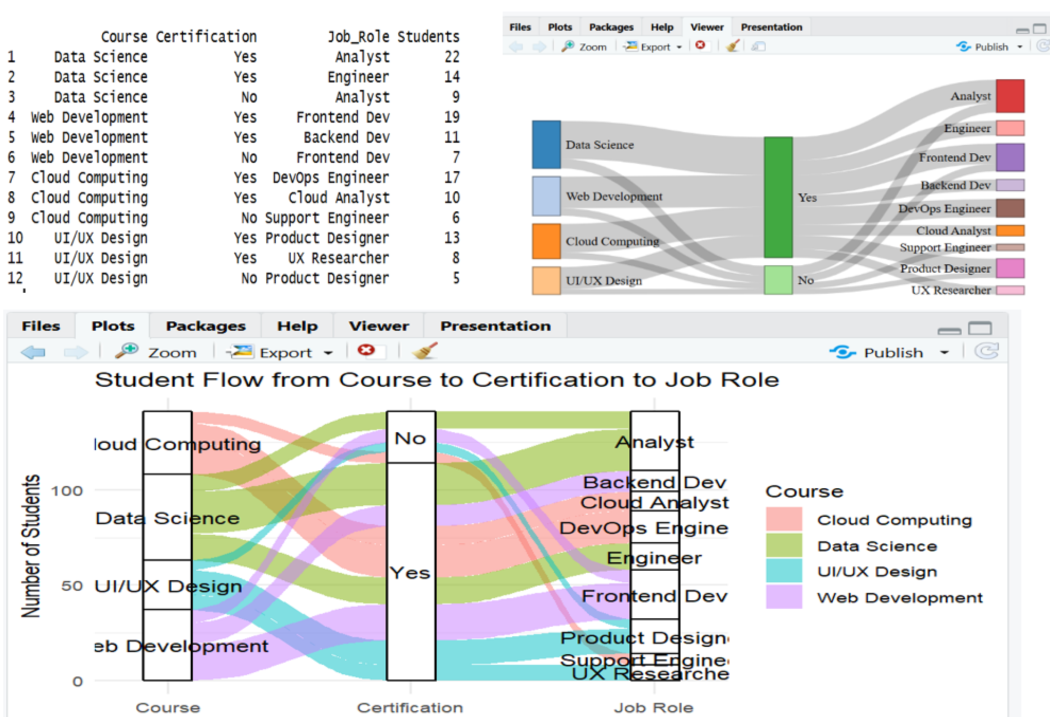
1. Create a Sankey diagram representing the data.

2. Identify at least four potential sources of confusion or misleading interpretation.

3. Redesign the Sankey diagram with meaningful labels, consistent flow representation, and an appropriate visual hierarchy.

4. Create a parallel sets alternative and compare the two designs.

5. Explain which visualization better communicates the major student career flows and why.



Flow-Based Proportions

- The data represents **Course → Certification → Job Role**.
- A Sankey diagram shows the movement of students between stages.
- Similar colors can make different flows difficult to distinguish.
- Unlabelled flows can hide the actual number of students.
- Flow crossings can create confusion.
- The redesigned Sankey uses clear labels and meaningful flow representation.
- A parallel sets plot is an alternative visualization.
- Sankey is better for showing **flow magnitude**.
- Parallel sets are better for comparing **categorical pathways** across all stages.